When blood reaches pulmonary capillaries, deoxyhemoglobin is converted into oxyhemoglobin. The hydrogen ions are released from hemoglobin because oxyhemoglobin has a weaker affinity for H^+ ions. Due to this, bicarbonate ions diffuse into erythrocytes which, in turn combine with H^+ ions and form carbonic acid. At low partial pressure of carbon dioxide, carbonic anhydrase converts carbonic acid into carbon dioxide and water within erythrocytes. Thus, bicarbonate ions reduce H^+ ions in lungs.

Role of Kidney

The kidney maintains pH by excreting either acidic or basic urine. Excretion of acidic urine increases pH whereas basic urine excretion decreases pH in extracellular fluid. Large amount of H^+ ions are secreted into the tubular lumen by tubular epithelial cells. If they are removed in urine, they will increase the pH of extracellular fluid. Large amount of bicarbonate ions are also continuously secreted in the renal tubules. If they are excreted in urine, then they reduce pH by retaining H^+ ions. If more hydrogen ions are removed than bicarbonate ions, then there will be a net loss of acid, whereas, if more bicarbonate ions are filtered than hydrogen ions are secreted, then there will be a net loss of base. These functions are achieved by mainly three components, namely, bicarbonate buffer system, phosphate buffer system and ammonia. The role of these components are as follows:

Bicarbonate buffer system in kidney

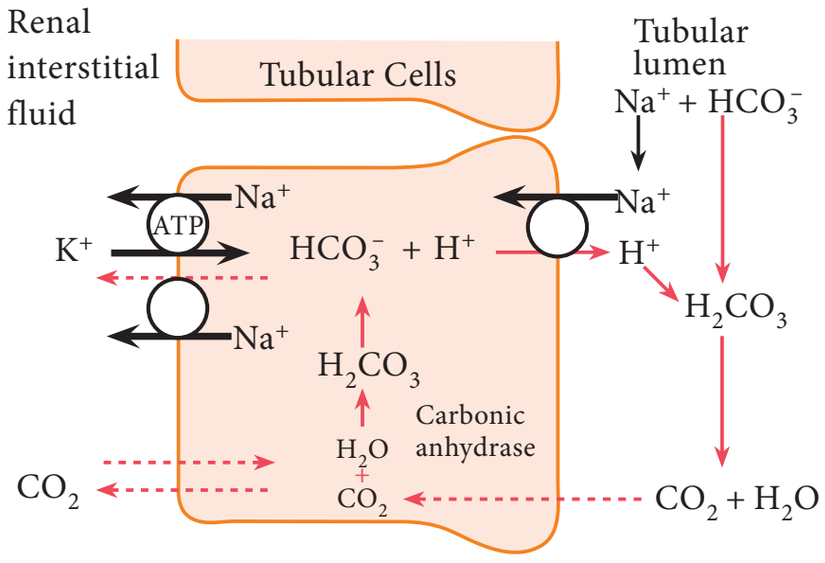

Figure 1.22 Role of bicarbonate ions in kidney tubule

The bicarbonate ions, freely filtered through glomerulus, combine with hydrogen ions and form carbonic acid which, in turn, dissociates into carbon dioxide and water. The carbon dioxide then diffuses into tubular cells where it again combines with water to form carbonic acid in the presence of carbonic anhydrase. Thus, bicarbonate and hydrogen ions are reabsorbed and retained. This pattern of H^+ ion secretion occurs in proximal convoluted tubule, ascending loop of Henle and early part of distal convoluted tubule.

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